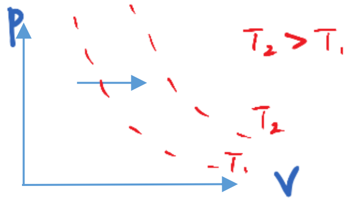
| | | |
|---|---|---|
| | | $T = 360\ K$ |
| 17 | D | Scenario 1: Positive work is done by gas at constant pressure □ Draw isotherms to visualize the process on a p-V diagram. Positive work done by gas indicates expansion of gas so the isobaric process will cause it to move from an isotherm of lower temperature, $T_1$, to an isotherm of a higher temperature, $T_2$.<br><br><br>Scenario 2: Gas is allowed to expand while heat is supplied slowly to it □ isothermal expansion is a <u>slow</u> process. Gas does positive work when heat supplied to it, resulting in no change in internal energy.<br>Scenario 3: Gas undergoes an adiabatic expansion □ expansion corresponds to positive work done by gas, hence negative work done on gas. Adiabatic process is one where no heat is exchanged with surrounding, hence Q=0. Applying the first law of thermodynamics, ΔU=Q+W=0+(negative value)=negative value (internal energy decreases) |